

SMERA RAWAL
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EDUCATION

Vellore Institute of Technology, Chennai (Pursuing) 2024 – 2028
B.Tech in Computer Science and Data Science
Relevant Coursework: Operating Systems, DBMS, OOPs, Data Structures, Computer Networks

TECHNICAL SKILLS

Languages: Python, C, C++, Java
Web: HTML, CSS, JavaScript
Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Feature Engineering, LSTM, CNN, Transformers, Neural Networks, GenAI, Reinforcement Learning, NLP
Libraries/Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras
Tools: Git, Linux, VS Code

RESEARCH AND EXPERIENCE

Learning Stellar Differential Rotation and Spot Evolution from Photometric Time Series (Ongoing)

- Investigating differential rotation and starspot evolution from photometric light curves using deep sequence modeling.
- Rotational modulation and spot migration via LSTM, 1D CNN and Transformers for long temporal dependencies.
- Applying Lomb-Scargle periodogram for irregular time-series frequency extraction and frequency-domain splitting.
- Designing sliding-window temporal representations and evaluating robustness under synthetic noise injection.
- Interpreting learned latent representations to study astrophysical rotational dynamics and variability patterns.

Quantitative Finance Research Consultant at WorldQuant

- Developed multiple alpha signals based on market microstructure and statistical patterns in financial time-series
- Achieved Maximum Sharpe Ratio of 2.66, returns of over 24 while maintaining a drawdown less than 5
- Applied statistical validation techniques including Sharpe ratio, IC and turnover analysis

Data Analytics Intern, Unessa Foundation

- Analyzed and processed datasets to extract actionable insights supporting organizational initiatives.
- Performed data cleaning, visualization, and reporting to improve decision-making efficiency.

PROJECTS

Astronomical Light Curve Classification using LSTM vs MLP vs 1D CNN

- Built an end-to-end stellar time-series pipeline with preprocessing, normalization and synthetic variability modeling.
- Implemented and compared LSTM, MLP and 1D CNN architectures for star classification performance benchmarking.
- Analyzed temporal dependency capture using multiple sliding-window configurations and feature scaling strategies.
- Injected synthetic Gaussian noise and conducted hyperparameter tuning to evaluate robustness.

ML-Based Drug Repurposing Pipeline (GNN + LLMs)

- Built an end-to-end pipeline combining GNN-based binding affinity prediction with Tanimoto similarity on ChEMBL
- Modeled molecules as graphs using RDKit and trained a GIN network to predict pChEMBL values
- Integrated LangChain for biomedical knowledge extraction and interpretable reasoning of drug-target interactions
- Generated ranked repurposing candidates with explainable scoring and visualization

Visual Financial Intelligence System (ViT + RAG)

- Multimodal financial analysis pipeline using Vision Transformer (ViT) to extract patterns from stock chart images.
- Retrieval-Augmented Generation (RAG) system using FAISS for semantic retrieval of financial knowledge.
- Interpretable trading insights using large language models (LLMs), improving explainability of technical analysis.

LEADERSHIP & ACTIVITIES

Logistics Lead, TEDx VITC

- Managed event logistics, vendor coordination and cross-functional execution for large-scale campus events.

Management and Outreach Member, Google Developer's Group

- Organized technical events, workshops and external collaborations to foster developer engagement.

Sponsorships and Outreach Member, Enactus and Business Innovation Community

- Contributed to ideation sessions, startup case discussions and innovation-driven problem solving initiatives.

Content Writer, Sahayatha Club

- Created engaging and informative content for social initiatives, increasing outreach and audience engagement.

CERTIFICATIONS

WorldQuant Gold - IQC 2026, Generative AI Certification - IBM, Machine Learning, Stanford University (Andrew Ng)

RELEVANT ACADEMIC INTERESTS

Scientific Machine Learning, Time-Series Modeling, Stellar Variability Analysis, Representation Learning, Robust Deep Sequence Models